SS-S-535a
JULY 25, 1958
SUPERSEDING
Int. Fed. Spec. SS–S–00535(GSA–FSS)
October 1, 1957, and
Fed. Spec. SS–B–611a
June 26, 1952

FEDERAL SPECIFICATION

SODIUM BORATE, DECAHYDRATE, TECHNICAL (BORAX)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — Sodium borate decahydrate (see 6.2) covered by this specification is intended to be used as a cleansing and stiffening agent in laundry work and for general purposes.

1.2 Classification.

1.2.1 *Types.*—Sodium borate decahydrate shall be furnished in the following types, as specified:

Type I. —Crystalline
Type II. —Granular
Type III.—Powdered

1.2.2 *Grade.*—Sodium borate decahydrate as specified herein, shall be furnished in one grade.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

RR–S–366—Sieves, Standard for Testing Purposes.

UU–P–271 — Paper, Wrapping, Waterproofed Kraft.

UU–S–48—Sacks, Paper Shipping.

PPP–B–35 — Bags: Textile, Shipping, Burlap, Cotton and Waterproof Laminated.

PPP–B–585—Boxes; Wood, Wirebound.

PPP–B–591—Boxes, Fiberboard, Wood-Cleated.

PPP–B–601—Boxes, Wood, Cleated-Plywood.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636—Boxes, Fiber.

PPP–T–60 — Tape; Pressure Sensitive Adhesive.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

FSC 6810

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–A–140 — Adhesive, Water-Resistant, Waterproof Barrier-Materials.

MIL–B–10377—Box, Wood, Cleated, Veneer, Paper Overlaid.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

ASTM Standards—Part 7

ASTM Designation D 501–55—Sampling and Chemical Analysis of Alkaline Detergents.

(Copies may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also widely distributed among technical groups and using Federal agencies.

2.2.1 The requirements of this specification include the requirements of the Standard Specifications for Borax, ASTM Designation: D 929, issued by the American Society for Testing Materials.

3. REQUIREMENTS

3.1 Assay.—The sodium borate decahydrate shall be not less than 99.5 percent as $Na_2B_4O_7.10H_2O$ when tested as specified in 4.4.3.2.

3.2 Free acid.—There shall be no free acid in the sodium borate decahydrate when tested as specified in 4.4.3.1.

3.3 Free alkali.—The free alkali shall be not more than 0.25 percent as NaOH when tested as specified in 4.4.3.1.

3.4 Water insoluble matter.—The sodium borate decahydrate shall contain not more than 0.1 percent water insoluble matter when tested as specified in 4.4.4.

3.5 Particle size (see 4.4.5).

3.5.1 Type I, crystalline sodium borate decahydrate shall be in the form of irregular shaped crystals, of such size that none shall be retained on a 1-inch sieve and the residue retained on a No. 8 sieve shall be not less than 90 percent.

3.5.2 Type II, granular sodium borate decahydrate shall be in the form of small crystals of such size that not more than 0.1 percent residue shall be retained on a No. 10 sieve and the residue retained on a No. 80 sieve shall be not less than 70 percent.

3.5.3 Type III, powdered sodium borate decahydrate shall be in the form of a fine white powder. Not more than 0.1 percent residue shall be retained on a No. 25 sieve and at least 12 percent shall pass through a No. 200 sieve.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Lot.*—For the purposes of sampling, a lot shall consist of all the sodium borate

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decahydrate of the same type, manufactured as one batch and offered for delivery at one time.

4.1.1.1 *Batch.*—A batch shall be that quantity of the material manufactured at the same time. When practical, all sodium borate decahydrate of the same type supplied on one order shall be from the same manufacturer's batch.

4.1.2 *Sampling for inspection of filled containers.*—A random sample of filled containers shall be taken in accordance with Military Standard MIL–STD–105 at Inspection Level I and Acceptance Quality Level (AQL) = 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.1.3 *Sampling for tests.*

4.1.3.1 *Packaged cartons.*—When packaged in cartons, one carton shall be taken at random from not less than 1 percent of the number of the vendor's packed shipping boxes provided such containers contain not less than 50 pounds each. In the case of smaller packed boxes, one shall be taken at random from each lot of boxes totaling not to exceed 5,000 pounds. The total sample shall in all cases consist of not less than three cartons taken at random. With very large lots, where the sample drawn as above will amount to more than 20 pounds, the percentage of packages sampled shall be reduced, so that the amount drawn shall not exceed 20 pounds. Wrap the individual carton tightly in paraffined paper at once and seal by rubbing the edges with a heated iron. The inspector should weigh each wrapped carton accurately; record its weight and the date of weighing on the wrapper; place the wrapped container in an airtight container which should be nearly filled; seal, mark, and send to the laboratory for test. Samples should be kept cool until tested. The seller shall have the option of being represented at the time of sampling and shall be furnished with a duplicate sample, if he so requests.

4.1.3.2 *Bags.*—When in bags, a grab sample of not less than 1 percent of the number of bags containing 100 pounds each, shall be taken. In case of 50-lb. bags, a grab sample of not less than one-half pound shall be taken at random from each lot of bags totaling not to exceed 10,000 pounds. The total sample shall in all cases consist of not less than three grab portions taken at random from separate bags. With very large lots, where the sample drawn as above will amount to more than 20 pounds, the percentage of bags sampled shall be reduced so that the amount drawn shall not exceed 20 pounds. The inspector should rapidly mix the sample, place in an airtight container, which he shall fill, seal, mark, weigh accurately, record its weight and date of weighing on package, and send to the laboratory for test. Samples should be kept cool until tested. The seller shall have the option of being represented at the time of sampling and shall be furnished with a duplicate sample, if he so requests.

4.2 Inspection.

4.2.1 *Inspection of filled containers.*—Each sample filled container shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings. Each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL–STD–105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or repaired all nonconforming containers.

4.3 Lot acceptance tests.—The sample specimens selected in accordance with 4.1.3 shall be subjected separately to the tests specified in 4.4. If any specimen fails one or more of these tests the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests

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provided the contractor has removed or reworked all nonconforming products.

4.4 Test procedures.

4.4.1 *Preparation of sample.* — Mix the sample rapidly; if desired, quarter down to about 1 pound, and weigh out all portions for analysis at once. Unused portions of the sample used for analysis shall be preserved in an airtight container in a cool place. Note the condition of the sample.

4.4.2 *Retest.*—When a determination shows nonconformity with specification, a duplicate shall be run.

4.4.3 *Chemical tests.*

4.4.3.1 *Total alkalinity or acidity.* — The total alkalinity or acidity shall be determined in accordance with ASTM Designation: D 501–55, section 38a. Calculations shall be made in accordance with paragraph 4.4.3.3.

4.4.3.2 *Assay.*—The total sodium borate decahydrate shall be determined in accordance with ASTM Designation: D 501–55, section 38b. Calculations shall be made in accordance with 4.4.3.3.

4.4.3.3 *Calculations.* — Calculate the percentages of sodium borate decahydrate and of free alkali or free acid, as follows:

Percent $Na_2B_4O_7.10H_2O - BN_B \times 9.540$

If AN_A is greater than BN_B,

Percent free alkali as NaOH $= (AN_A - BN_B) \times 2.000$

If AN_A is less than BN_B,

Percent free acid as $H_3BO_3 = (BN_B - AN_A) \times 6.184$

where: B = milliliters of sodium hydroxide solution required for titration in 4.4.3.2,

N_B = normality of the sodium hydroxide solution,

A = milliliters of hydrochloric acid required for the titration in 4.4.3.1, and

N_A = normality of the hydrochloric acid.

4.4.3.4 *Explanatory notes.*—In sodium borate decahydrate (free from excess B_2O_3 or Na_2O), the acid titration is half the subsequent alkali titration (factor, acid to sodium borate decahydrate = 0.1908, alkali to sodium borate decahydrate = 0.0954). If acid titration exceeds this proportion, alkali other than that combined with sodium borate decahydrate is indicated; if the alkali titration is greater than twice the acid titration, free boric acid is indicated. The mannitol or glycerol should be made neutral with 0.1*N* NaOH before use, in case it contains free fatty acid. Sodium borate decahydrate ($Na_2B_4O_7.10H_2O$) effloresces in warm, dry air. Hence, a sample that has lost some of its water of crystallization will show high results when calculated to $Na_2B_4O_7.10H_2O$.

4.4.4 *Water insoluble matter.*—The water insoluble matter shall be determined in accordance with ASTM Designation: D 501–55, section 39.

4.4.5 *Sieve test.*—This test should be carried out with sieves conforming to Federal Specification RR–S–366.

4.4.5.1 *Type I crystalline borax.*—Transfer a 1-pound sample to a 1-inch sieve and sift, tapping the sieve frame from time to time and catching the residue retained on a No. 8 sieve. The residue shall be not less than 90 percent of the original weight of the sample. Type I material should be tested before packaging.

4.4.5.2 *Type II granular borax.*—Transfer 100 gm. of the well-mixed sample, without previous drying to a dry No. 10 sieve and sift, tapping the sieve frame from time to time or alternatively use a mechanical sieve shaker for 10 minutes and catching all of the material passing through the No. 10 sieve on a dry No. 80 sieve. The sifting on the No. 10 sieve is continued until the weight of the residue is not reduced by more than 0.1 gm.

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on further sifting for 1 minute, tapping the sieve frame as before. Calculate the final weight of residue to percentage retained on the No. 10 sieve. Sift the material on the No. 80 sieve, tapping the frame from time to time, until the weight of the residue on the No. 80 sieve is not reduced by more than 0.1 gm. after further sifting for 1 minute, tapping the frame as before. Add the final weight of the residue retained on the No. 10 sieve to the final weight of the residue retained on the No. 80 sieve and calculate the sum to percentage of residue retained on the No. 80 sieve.

4.4.5.3 *Type III powdered borax.* — The sieve test for type III is carried out in essentially the same manner as the test described under 4.4.5.2 except that No. 25 sieve and No. 200 sieve are used in this test and the percentages to be determined are the residue retained on the No. 25 sieve and the powder which passes through the No. 200 sieve.

5. PREPARATION FOR DELIVERY

For civilian agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Levels A and C.*

5.1.1.1 *One or five pounds.*—Sodium borate decahydrate shall be packaged in commercial cartons in quantities as specified. The cartons shall be wrapped in water-proofed flexible barrier materials conforming to Federal Specification UU–P–271, class C1 or C2, with all folds, seams, and laps sealed with water-resistant adhesive conforming to Military Specification MIL–A–140, type II, grade A.

5.1.1.2 *Fifty or one hundred pounds.*— Sodium borate decahydrate shall be packaged in quantities as specified, in laminated textile bags conforming to type III of Federal Specification PPP–B–35, or in multiwall paper shipping sacks conforming to Federal Specification UU–S–48.

5.2 Packing.

5.2.1 *Level A.* — Sodium borate decahydrate, packaged as specified in 5.1.1, shall be packed in cleated-plywood, nailed wood, or wirebound wood boxes conforming to Federal Specification PPP–B–601, PPP–B–621, and PPP–B–585, respectively, or fiber boxes conforming to PPP–B–636, type I or II, class 3. The waterproof bag will not be required when the container conforms to Federal Specification PPP–B–636 and is sealed at all joints and seams, including manufacturer's joints with tape conforming to Federal Specification PPP–T–60. The gross weight shall not exceed approximately 200 pounds. Contents shall be enclosed within a sealed waterproof bag or its equivalent made from material conforming to Federal Specification UU–P–271, for case-liner material. Seams and closures shall be sealed with adhesive conforming to Military Specification MIL–A–140.

5.2.1.1 No overpacking of sodium borate decahydrate in 50 or 100 pound quantities will be required.

5.2.2 *Level B.*—Sodium borate decahydrate in 1 or 5 pound quantities packaged as specified in 5.1.1 shall be packed in wood-cleated fiberboard, cleated-plywood, nailed and lock-corner wood, wirebound wood, corrugated or solid fiberboard, or wood cleated paper overlaid boxes conforming to Federal Specification PPP–B–591, PPP–B–601, PPP–B–621, PPP–B–585, PPP–B–636, or Military Specification MIL–B–10377, respectively, at the option of the contractor. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitation and shall conform to the special requirements of the applicable box specification.

5.2.2.1 No overpacking of sodium borate decahydrate in 50 or 100 pound quantities will be required.

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5.2.3 *Level C.*—Unless otherwise specified, the subject commodity shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to the point of delivery. Shipping containers shall show no indication of leakage on delivery.

5.3 **Marking.**

5.3.1 *Civil agencies.* — Interior packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.* — In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein and should specify the following in procurement documents:

(*a*) Number, title, and date of this specification.

(*b*) Total weight required and size or weight of individual containers.

(*c*) Type as specified in 1.2.1.

(*d*) Level of packaging and packing required (see 5.1 and 5.2).

6.2 Sodium borate decahydrate (Borax) is also known as sodium tetraborate and sodium pyroborate.

6.3 **Other grades.**—The American Chemical Society specifications for sodium borate covers a grade suitable for use as a reagent. The U. S. Pharmacopoeia specification covers a grade suitable for medicinal purposes.

6.4 **Transportation description.**—Transportation description applicable to this item is:

Borax (sodium borate), less carload, in bags, barrels, boxes or steel parts; carload, in bulk or in packages.
Carload weight 36,000 pounds.

Borax (sodium borate), less truckload, in bags, barrels, boxes or steel parts; truckload, in bulk or in packages.
Truckload weight 36,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
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